

AERONEXUS TECHNOLOGIES

Incident Management Standard Operating Procedure (SOP)

Enterprise Operations Framework: Incident Classification, Escalation Matrices,
and Post-Mortem Protocols

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Cloud Operations

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Advanced Core

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Operational Mandate

Compliance with this Standard Operating Procedure is mandatory for all members of the engineering, infrastructure, security, and application support divisions. Deviations from the communication intervals or timeline commitments established herein during an active P1/P2 event will trigger an automatic operational compliance review.

1.0 FUNCTIONAL SCOPE, PURPOSE, AND TAXONOMY

1.1 Engineering Context

AeroNexus Technologies maintains automated high-throughput data engines and distributed service layers supporting critical financial and logistics enterprises globally. System downtime or performance degradation impacts client transactions directly. This document establishes the definitive operational protocol for identifying, triaging, stabilizing, and conducting root-cause investigations for all unexpected infrastructure anomalies, software regressions, and security incidents.

1.2 Boundaries of Application

This SOP governs the behavior of the Global Site Reliability Engineering (SRE), Network Operations Center (NOC), Infrastructure Platform, Cybersecurity Operations (SecOps), and Application Support teams. It addresses all service disruptions occurring within the *.aeronexus.net public production perimeters, multi-cloud hosting environments (AWS, Azure), and internal infrastructure blocks.

1.3 Core Definitions & Tooling Context

- **ServiceNow ITSM:** The authoritative system of record for all corporate Configuration Items (CIs), Change Management Tickets, and formal Incident Records.
- **PagerDuty:** The dynamic alerting and real-time on-call scheduling engine responsible for translating automated telemetry anomalies into engineer pages.
- **Incident Commander (IC):** The individual responsible for directing the technical investigation and mitigation efforts during a major incident. The IC holds absolute authority over system changes, rollbacks, and operational overrides for the duration of the incident.
- **Blast Radius:** The measurable scope of business impact, including affected tenant IDs, data throughput degradation, or API transaction drop rates.

2.0 INCIDENT SEVERITY MATRIX & PRIORITY CATEGORIZATIONS

2.1 Defining Severity vs. Priority

AeroNexus differentiates between architectural Severity (the degree of systems degradation) and operational Priority (the speed at which remediation teams must react). Calculations are executed within ServiceNow based on a combined Impact and Urgency cross-reference table.

Priority Level	Classification Type	Technical Impact Definition Threshold Criteria
P1 - Critical	Major Incident (MI)	Complete failure of a core production application. >35% of global users impacted, or direct leakage of client data perimeters. No known workaround available.
P2 - High	Significant Disruption	Partial degradation of core architectures (e.g., ProductX API latency exceeds 1500ms). Substantial segment of users impacted (>15%). Redundancy systems failing.
P3 - Moderate	Standard Incident	Minor system functionality lost, but core application workflows remain viable. Impact isolated to specific customer tenants or regional office spaces. Workaround exists.
P4 - Low	Cosmetic / Minor	Non-blocking software issues, documentation errors, or reporting anomalies. No direct customer transaction disruption.

2.2 Automatic Downgrade Restrictions

Once an incident ticket has been flagged as a P1 or P2 within the system, its priority status cannot be manually adjusted downward by a support engineer without direct authentication from the acting Incident Commander. This prevents artificial SLA manipulation across the operational queue.

3.0 OPERATIONAL RESPONSE TIMELINES & SLAS

3.1 Service Level Agreements (SLAs)

Remediation timelines are systematically monitored via automated ServiceNow tracking loops. Failure to fulfill intermediate milestones automatically triggers pages up the management chain to ensure appropriate engineering resources are deployed.

Priority	Triage & Acknowledgment	Technical Assessment	Internal Comms Cadence	Target Workaround / Resolution
P1	≤ 5 Minutes	≤ 15 Minutes	Every 20 Minutes	≤ 2 Hours
P2	≤ 15 Minutes	≤ 45 Minutes	Every 60 Minutes	≤ 6 Hours
P3	≤ 2 Hours	≤ 8 Hours	Every 24 Hours	≤ 48 Hours
P4	≤ 24 Hours	≤ 72 Hours	Upon State Change	≤ 10 Business Days

3.2 Acknowledgment Definitions

Acknowledgment is defined as an active engineer claiming the PagerDuty alert thread, updating the ServiceNow record to "In Progress," and entering the designated Slack incident communications channel. Passive automated host responses do not satisfy triage criteria.

SLA Breach Mitigation Actions

If an active P1 incident ticket approaches 75% of its target resolution timeline (90 minutes from initial tracking check) without achieving system stabilization, the system automatically page-notifies the VP of Cloud Operations and the Chief Technology Officer to unlock external vendor diagnostic pipelines.

4.0 DYNAMIC TECHNICAL & MANAGEMENT ESCALATION WORKFLOWS

4.1 On-Call Rotation Escalation Steps

When an automated monitoring alert passes through PagerDuty to the primary on-call engineer, the individual has exactly 5 minutes to acknowledge the page before the platform executes an automated escalation sequence.

1. **Primary On-Call Notification:** Initial page via mobile app, SMS, and automated voice interface call.
2. **Secondary On-Call (Level 2):** Triggered automatically if the primary engineer fails to respond within 5 minutes. The secondary engineer assumes active ownership of the incident track immediately.
3. **Engineering Manager Escalation:** Triggered if both primary and secondary fail to acknowledge within 10 total minutes of incident discovery. The manager must clear operational blockers and assign resources manually.
4. **Division Director / VP Paging:** Triggered at 15 minutes of non-response, indicating a major support pipeline failure. This triggers an instant compliance audit record.

4.2 Cross-Functional Engagement Mapping

Major infrastructure anomalies frequently require intervention from adjacent technical groups. The table below outlines cross-functional engagement triggers:

Technical Anomaly Context	Target Team Engagement	Required System Action Protocol
Database Lockups / Query Blocks	DBA Platform Core	Trigger secondary read-replica promotion path.
BGP Routing Drop / DDoS Attack	Network Edge Engineering	Activate cloud scrubbers; redirect traffic paths.
Unauthorized IAM Token Generation	SecOps Threat Intel	Isolate credentials; initiate host forensic capture.

5.0 MAJOR INCIDENT ACTIVATION & MAJOR INCIDENT WAR ROOM PROTOCOLS

5.1 Declaration and Initialization

A Major Incident (MI) is declared immediately upon verification of P1 criteria, or when a P2 event fails to respond to containment workflows within 30 minutes of deployment. Declaration must be formally executed via the ServiceNow interface by the NOC lead, SRE director, or any designated Incident Commander.

5.2 War Room Operational Roles

Upon MI declaration, an automated bridge link is generated, and the mandatory communications lines are initialized. The following three structural roles must be explicitly filled immediately:

- **Incident Commander (IC):** Drives the technical investigation, directs engineering tasks, maintains structural focus on mitigation, and approves emergency code modifications or configuration changes.
- **Communications Lead (CL):** Manages the incident timeline, maintains the internal ticket log, and drafts customer-facing status updates. The CL isolates the IC from external status requests.
- **Operations Liaison (OL):** Coordinates with Customer Success leads, Account Directors, and Legal teams, translating technical assessments into business impact language.

5.3 Strict Bridge Etiquette

The MI War Room bridge is a structured technical workspace. The following rules are enforced by the IC:

1. Only actively assigned engineers and operations leads may speak on the bridge. Observational personnel must remain muted.
2. Speculative or non-data-driven assertions are prohibited. All technical assertions must be backed by live metric dashboard views or log extracts.
3. When the IC commands a system quiet window ("Hold for Diagnostics"), all network changes and debugging activities must cease instantly.

6.0 POST-MORTEM ARCHITECTURE & PROBLEM MANAGEMENT RECORDS

6.1 Blameless Culture & Timeline Reconstruction

AeroNexus Technologies operates on a ****Strictly Blameless Post-Mortem Doctrine****. The core objective of any post-incident review is to discover architectural vulnerabilities, process blind spots, and tooling deficiencies, rather than assign individual faults. Every P1 and P2 event requires a formal Post-Incident Review (PIR) within the ServiceNow Problem Management engine.

6.2 Post-Mortem Delivery Timelines

- **Preliminary Post-Mortem:** A rough technical summary and timeline reconstruction must be drafted and shared within 24 hours of incident resolution.
- **Formal Post-Mortem Document:** The complete document, including comprehensive log verification, root cause analysis, and permanent remediation actions, must be finalized within 5 business days.

6.3 Action Item Tracking & Governance

Remediation tasks generated during the PIR are codified into system tickets and governed by strict completion windows based on safety priority:

Action Item Priority Classification	Target Resolution Window	Governance Compliance Mandate
Critical (P0 Task)	≤ 7 Calendar Days	Blocks subsequent feature releases for the target system.
High (P1 Task)	≤ 30 Calendar Days	Requires director-level sign-off if deferred.
Medium (P2 Task)	≤ 90 Calendar Days	Tracked in team velocity sprints.

7.0 INCIDENT SIMULATIONS, ACTIONABLE FAQs, AND GOVERNANCE

LIVE INCIDENT SCENARIO: PRODUCTX DATABASE CONNECTION EXHAUSTION

At 14:22 UTC, automated Datadog metrics flag a sudden spike in HTTP 500 error counts on the ProductX ingestion layer. PagerDuty notifies the SRE primary engineer. The engineer logs into the system at 14:25 UTC and triage confirms database connection pool exhaustion caused by an unindexed query rollout. Following the SOP, a P1 is declared, a war room bridge is activated, and the IC orders an immediate rollback of the change set. Service returns to normal operations at 14:45 UTC.

7.2 Frequently Asked Questions (FAQs)

Q1: Can a customer success manager request the activation of a Major Incident War Room bridge for an issue impacting only one customer account?

A: Generally, no. War rooms are reserved for P1/P2 events impacting multiple clients or core systems. However, if that single account belongs to a Premier Strategic Tier client and directly breaks their primary transaction path, the Operations Liaison may grant an exception and initiate an expedited bridge loop.

Q2: What should an engineer do if they discover an active security vulnerability while debugging a live production network issue?

A: The engineer must immediately inform the Incident Commander, who will pivot the incident pipeline to include SecOps on a secure communication sub-channel. The incident record will be flagged as "Restricted Confidentiality" within ServiceNow to limit access to authorized security response personnel only.

7.3 SOP Maintenance and Drills

This document is owned by the SRE Operations Board and undergo comprehensive revision cycles every 12 months. To maintain high readiness, the organization conducts unannounced "Chaos Engineering" simulations and tabletop disaster recovery drills quarterly.